

```

<program> ::= <statement> <statements>
<statements> ::= <statement> <statements> | epsilon
<statement> ::= <assignment> | <list_expression> | <if_statement> | <for_loop> |
<print_statement>
<assignment> ::= <identifier> "=" <list_expression>
<list_expression> ::= <list_creation> | <list_expression> <filter_op> | <list_expression>
<add_op> | <list_expression> <remove_op> | <list_expression> <select_op>
<list_creation> ::= "List" "(" <record_list> ")"
<filter_op> ::= "?" <condition>
<condition> ::= <simple_condition> | <condition> "and" <condition> | "(" <condition> ")"
<simple_condition> ::= <identifier> <operator> <value>
<add_op> ::= "+" <identifier> "=" <value>
<remove_op> ::= "-" <identifier>
<select_op> ::= "->" "(" <identifier_list> ")"
<identifier_list> ::= <identifier> <more_identifiers>
<more_identifiers> ::= "," <identifier> <more_identifiers> | epsilon
<operator> ::= "==" | "!=" | ">" | "<" | ">=" | "<="
<value> ::= <string> | <number>
<string> ::= "" <characters> ""
<characters> ::= <character> <characters> | epsilon
<character> ::= <letter> | <digit>
<identifier> ::= <letter> <identifier_rest>
<identifier_rest> ::= <letter_or_digit> <identifier_rest> | epsilon
<number> ::= "0" | <optional_minus> <non_zero_digit> <digits>
<optional_minus> ::= "-" | epsilon
<digits> ::= <digit> <digits> | epsilon
<letter> ::= "a" | ... | "z" | "A" | ... | "Z"
<digit> ::= "0" | "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9"
<non_zero_digit> ::= "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9"
<record_list> ::= "[" <record_list_content> "]"
<record_list_content> ::= <record> <more_records> | epsilon
<more_records> ::= "," <record> <more_records> | epsilon
<record> ::= "{" <pair> <more_pairs> "}"
<more_pairs> ::= "," <pair> <more_pairs> | epsilon
<pair> ::= <identifier> ":" <value>
<if_statement> ::= "if" <condition> <statement_block> <optional_else> "end"
<optional_else> ::= "else" <statement_block> | epsilon
<statement_block> ::= <statement> <statements>
<for_loop> ::= "for" <identifier> "in" <list_expression> <statement_block> "end"
<print_statement> ::= "puts" <list_expression>

```